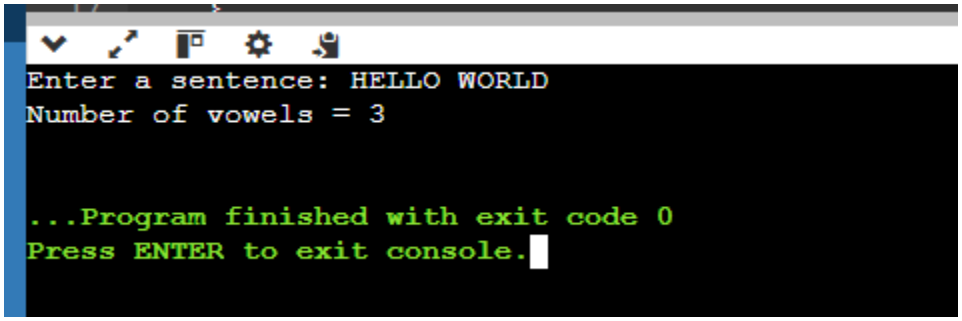
EXPERIMENTS

21. In a class, an English teacher was teaching the vowels and consonants to the students. She says “Vowel sounds allow the air to flow freely, causing the chin to drop noticeably, whilst consonant sounds are produced by restricting the air flow”. As a class activity the students are asked to identify the vowels and consonants in the given word/sentence and count the number of elements in each. Write an algorithm to help the student to count the number of vowels in the given sentence.
CODE:

```
#include <stdio.h>

int main() { char str[200]; int i, count = 0;

printf("Enter a sentence: ");
fgets(str, sizeof(str), stdin);

for (i = 0; str[i] != '\0'; i++)
{
   if (str[i] == 'a' || str[i] == 'e' ||
      str[i] == 'i' || str[i] == 'o' ||
      str[i] == 'u' || str[i] == 'A' ||
      str[i] == 'E' || str[i] == 'I' ||
      str[i] == 'O' || str[i] == 'U')
   {
      count++;
   }
}

printf("Number of vowels = %d\n", count);

return 0;


}
```

OUTPUT:



22. Write a LEX program to print all HTML tags in the input file.
Input Source Program: (sample.html)
<html>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
CODE:

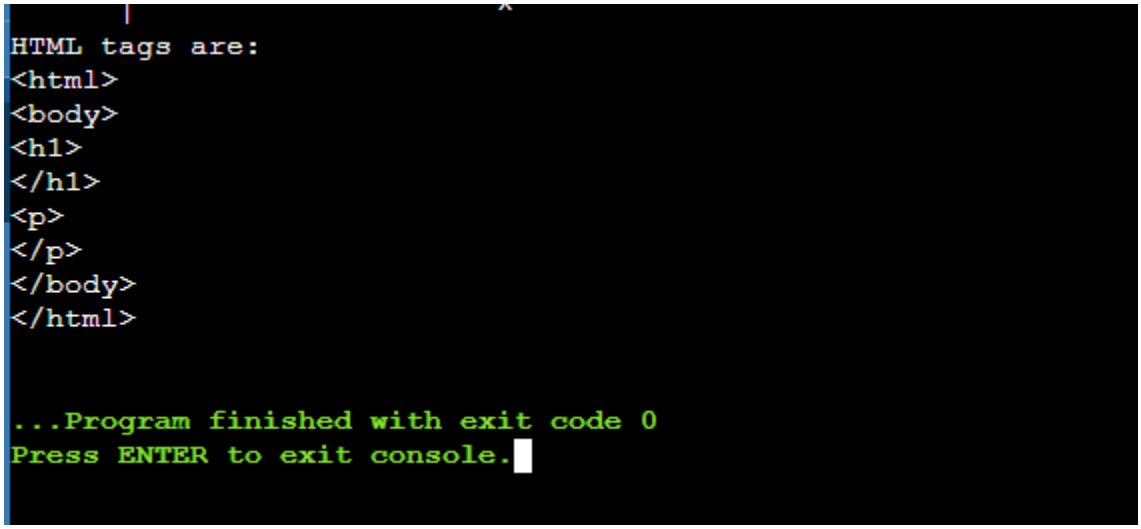
#include <stdio.h> #include <string.h>

int main() { char html[] = "<html>\n" "<body>\n" "<h1>My First Heading</h1>\n" " My first paragraph. \n" "</body>\n" "</html>";

int i, j;

printf("HTML tags are:\n");

```
for (i = 0; html[i] != '\0'; i++)
{
   if (html[i] == '<')
   {
      j = i;

      while (html[j] != '\0' && html[j] != '>')
         j++;

      if (html[j] == '>')
      {
         for (; i <= j; i++)
            putchar(html[i]);

         printf("\n");
```

```
    }
  }
}

return 0;


}
```

OUTPUT: